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JavaScript Numbers Cheatsheet



Number constants

Constant	Value	Description
<code>Number.EPSILON</code>	2.220446049250313e-16	Smallest difference between two representable numbers.
<code>Number.MAX_VALUE</code>	1.7976931348623157e+308	Largest finite number.
<code>Number.MIN_VALUE</code>	5e-324	Smallest positive number.
<code>Number.MAX_SAFE_INTEGER</code>	9007199254740991	Largest safe integer.
<code>Number.MIN_SAFE_INTEGER</code>	-9007199254740991	Smallest safe integer.
<code>Number.POSITIVE_INFINITY</code>	Infinity	Positive infinity.
<code>Number.NEGATIVE_INFINITY</code>	-Infinity	Negative infinity.
<code>Number.NaN</code>	NaN	Not-a-Number.

Number methods

Method	Description
<code>Number.isNaN(value)</code>	Returns <code>true</code> if <code>value</code> is <code>NaN</code> .
<code>Number.isFinite(value)</code>	Returns <code>true</code> if <code>value</code> is a finite number.
<code>Number.isInteger(value)</code>	Returns <code>true</code> if <code>value</code> is an integer.
<code>Number.isSafeInteger(value)</code>	Returns <code>true</code> if <code>value</code> is a safe integer.
<code>Number.parseFloat(string)</code>	Parses a string and returns a floating-point number.
<code>Number.parseInt(string, radix)</code>	Parses a string and returns an integer.
<code>Number.toFixed(digits)</code>	Returns a string with a fixed number of decimals.
<code>Number.toPrecision(precision)</code>	Returns a string with a specified precision.
<code>Number.toString(radix)</code>	Returns a string representation in the given base.
<code>Number.toExponential(fraction)</code>	Returns a string in exponential notation.
<code>Number.valueOf()</code>	Returns the primitive value of a Number object.

Parsing numbers

`Number.parseInt()`

Parses a string and returns an integer. Always specify the base to avoid unexpected results.

js



```
1 Number.parseInt("42", 10); // 42
2 Number.parseInt("ff", 16); // 255
3 Number.parseInt("10px"); // 10
```

`Number.parseFloat()`

Parses a string and returns a floating-point number.

js



```
1 Number.parseFloat("3.14"); // 3.14
2 Number.parseFloat("10.5px"); // 10.5
```

Number()

Converts a value to a number. Whitespace is ignored.

js



```
1 Number(" 100 "); // 100
2 Number("3.14"); // 3.14
3 Number(""); // 0
4 Number("hello"); // NaN
```

+ (unary plus)

The unary plus operator converts a string to a number. Prefer `Number()` for clarity.

js



```
1 +"99"; // 99
2 +"3.14"; // 3.14
3 +true; // 1
```

Formatting numbers

`toFixed()`

Returns a string with a fixed number of decimal places.

js



```
1 (1.2345).toFixed(2); // "1.23"  
2 (10).toFixed(4); // "10.0000"
```

`toPrecision()`

Returns a string with a specified total number of significant digits.

js



```
1 (1.2345).toPrecision(3); // "1.23"  
2 (1234).toPrecision(2); // "1.2e+3"
```

toString()

Returns a string representation in the given base.

js



```
1 (255).toString(16); // "ff"
2 (8).toString(2); // "1000"
```

toLocaleString()

Returns a language-sensitive formatted string.

js



```
1 (1234567).toLocaleString("en-US"); // "1,234,567"
2 (1234.5).toLocaleString("de-DE"); // "1.234,5"
```

toExponential()

Returns a string in exponential notation.

js



```
1 (12345).toExponential(2); // "1.23e+4"
```

Math helpers

Math.round()

Rounds to the nearest integer.

js



```
1 Math.round(4.4); // 4
2 Math.round(4.5); // 5
```

Math.floor()

Rounds down to the nearest integer.

js



```
1 Math.floor(4.9); // 4
2 Math.floor(-4.1); // -5
```

Math.ceil()

Rounds up to the nearest integer.

js



```
1 Math.ceil(4.1); // 5
2 Math.ceil(-4.9); // -4
```

Math.trunc()

Removes the decimal part.

js



```
1 Math.trunc(4.9); // 4
2 Math.trunc(-4.9); // -4
```

Math.abs()

Returns the absolute value.

js



```
1 Math.abs(-5); // 5
```

Math.max() / **Math.min()**

Return the largest or smallest of the given values.

js



```
1 Math.max(1, 5, 3); // 5
2 Math.min(1, 5, 3); // 1
```

`Math.random()`

Returns a random number from 0 (inclusive) to 1 (exclusive).

js



```
1 Math.random(); // e.g., 0.374829
```

`Math.pow()` / `Math.sqrt()`

Exponentiation and square root.

js



```
1 Math.pow(2, 3); // 2^3 -> 8
2 Math.sqrt(16); // 4
```

Special values

```
0.1 + 0.2 === 0.3
```

Returns `false` due to floating-point precision. Use rounding or `Number.EPSILON` for comparisons.

```
Number.isNaN(NaN)
```

Returns `true`. Always use `Number.isNaN` instead of `===` because `NaN === NaN` is `false`.

```
Number.isFinite(Infinity)
```

Returns `false`. `Infinity` and `-Infinity` are not finite numbers.

```
Number.isInteger(42)
```

Returns `true` for whole numbers without a fractional part.

```
Number.isSafeInteger(9007199254740992)
```

Returns `false`. Safe integers are those that can be exactly represented as a double.